

Executive Business Report

RetailIQ – AI-Powered Sales Forecasting & Demand Intelligence

Prepared for: Head of Supply Chain & Chief Financial Officer (CFO)

Prepared By: Gauri Lad

Executive Summary

RetailIQ is a business intelligence solution developed to help retail organizations understand historical sales performance, forecast future demand, detect unusual sales behavior, and identify high-demand product segments. The dashboard combines historical sales analysis with machine learning forecasting techniques to support data-driven inventory and supply chain decisions. By identifying demand trends and potential business risks early, RetailIQ enables better stock planning, minimizes inventory costs, and improves product availability. The system provides an easy-to-understand dashboard that allows management to monitor business performance without requiring technical expertise.

Key Findings from Sales Analysis (EDA)

The exploratory data analysis revealed several important business insights from historical retail sales.

Sales varied significantly across different months, indicating strong seasonal demand patterns. Certain months consistently generated higher revenue than others, suggesting that promotional campaigns and inventory planning should be aligned with these peak periods.

Regional analysis showed that the **West region generated the highest overall sales**, followed closely by the East region, while the South region contributed the lowest sales. This indicates that inventory allocation and marketing investments can be optimized based on regional demand.

Category analysis demonstrated that **Technology products contributed the largest share of revenue**, followed by Furniture and Office Supplies. Technology products therefore represent the strongest revenue-driving category and should receive higher inventory priority.

Overall, historical sales data indicates that customer demand is not uniform across time, regions, or product categories, making predictive analytics valuable for business planning.

Three-Month Sales Forecast

Three forecasting models—SARIMA, Prophet, and XGBoost—were evaluated to estimate future sales.

Model comparison showed that **XGBoost achieved the lowest forecasting error (lowest MAPE)**, making it the most reliable model for predicting future sales trends.

The forecast indicates that overall sales are expected to remain relatively stable during the next three months, with moderate month-to-month variation rather than sharp increases or declines. No evidence of severe business downturn or excessive growth was observed.

Forecast Confidence

The forecast should be considered **moderately reliable** because:

- Historical sales data follows recognizable patterns.

- Multiple forecasting models produced similar trends.
- XGBoost demonstrated the highest prediction accuracy among all tested models.

Business managers can therefore use these forecasts confidently for short-term inventory planning while continuing to monitor market conditions.

Top Three Anomalies Detected

The anomaly detection module identified unusual sales observations that differed significantly from normal business patterns.

1. Sudden High Sales Spike

A few transactions showed exceptionally high sales compared to surrounding periods. These are likely caused by seasonal promotions, festival discounts, or bulk customer purchases.

2. Unexpected Sales Drop

Some dates experienced unusually low sales. Possible causes include reduced customer demand, stock shortages, holidays, or temporary operational disruptions.

3. Irregular Demand Fluctuation

Certain products displayed inconsistent purchasing behavior that did not match historical trends. These anomalies may indicate changing customer preferences or external market influences.

Monitoring these unusual events enables businesses to respond proactively before they affect profitability.

Product Demand Segmentation & Inventory Strategy

Demand segmentation grouped products based on purchasing frequency and sales performance.

High-Demand Products

These products generate consistent sales and should maintain higher inventory levels to prevent stock shortages.

Recommended Strategy:

- Maintain sufficient stock
 - Prioritize supplier availability
 - Use automatic inventory replenishment
-

Medium-Demand Products

These products show moderate but stable demand.

Recommended Strategy:

- Maintain balanced inventory
- Review demand monthly
- Optimize reorder quantities

Low-Demand Products

These products sell infrequently and occupy warehouse space.

Recommended Strategy:

- Reduce inventory levels
 - Purchase only when required
 - Consider promotional discounts or product discontinuation
-

Business Recommendations

1. Prioritize High-Demand Technology Products

Technology products contribute the largest share of sales. Increasing inventory availability for these products will reduce stock-outs and maximize revenue opportunities.

Data Support: Technology category recorded the highest sales contribution during analysis.

2. Adopt XGBoost for Demand Forecasting

Among all forecasting models, XGBoost achieved the lowest Mean Absolute Percentage Error (MAPE), indicating superior forecasting accuracy.

Data Support: Model comparison identified XGBoost as the best-performing forecasting model.

3. Monitor Sales Anomalies Regularly

Unexpected spikes and drops in sales should be reviewed immediately to determine whether they result from promotional events, supply issues, or changing customer behavior.

Data Support: The anomaly detection module successfully identified abnormal sales patterns in the historical dataset.

Risk / Limitation

The forecasting models rely entirely on historical sales data. Unexpected external events such as economic changes, competitor activities, supply chain disruptions, or customer preference shifts cannot be predicted accurately using historical information alone.

Therefore, forecast results should be combined with business knowledge and current market conditions before making strategic decisions.

Conclusion

RetailIQ demonstrates how data analytics and machine learning can support smarter retail decision-making. The system successfully combines sales analysis, demand forecasting, anomaly detection, and product segmentation into a single interactive dashboard. By using these insights, retail businesses can improve inventory planning, reduce operational costs,

respond quickly to changing demand, and make informed business decisions. Overall, RetailIQ provides a practical and scalable solution for modern retail demand intelligence.